ComfyUI + MiniMax H3: what everything actually does

Plain English. No definitions that define nothing. Every part and setting gets the same three answers.

What is it?
In words, not more acronyms.

What does it do?
To your video, your sound, or your wait.

What do I set it to?
A real starting value for this machine.

Sized for `RTX 5060 Ti` `16 GB VRAM` `32 GB RAM`

1 The big picture READ THIS FIRST

H3 does not film anything. It starts with a screen of pure static and cleans it up a little at a time until it looks and sounds like your prompt. Picture and sound get cleaned up together in the same pass, which is why lips match words.

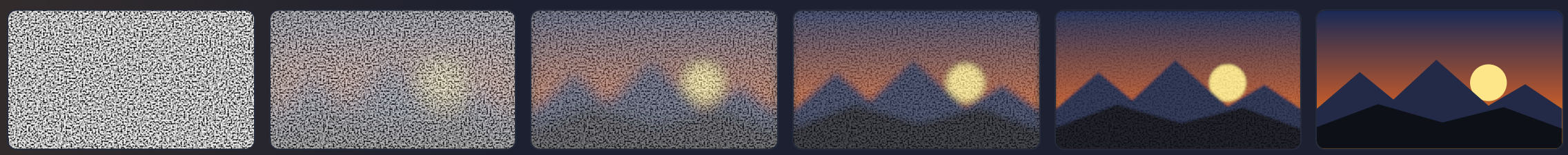
YOU
Prompt
Your words, plus pictures, clips or audio if the mode uses them.

THE READER
Text encoder
Reads the prompt, looks at your pictures, and turns it all into numbers the artist can follow. Runs once per prompt.
`Load CLIP`

THE ARTIST
Diffusion model
Starts from static and cleans it up one step at a time, steered by the reader's numbers. Nearly all of your wait is here.
`Load Diffusion Model`

THE UNPACKERS
Video VAE + Audio VAE
The artist works on a tiny compressed version. These turn it back into real frames and real sound.
`Load VAE, twice`

RESULT
MP4
24 frames per second, with stereo sound.



`start: static` `step 4` `step 8` `step 12` `step 16` `step 20: done`

A step is one cleanup pass. More passes, cleaner result, longer wait. (A drawing of the idea, not real output.)

Everything else on this page is one of three things: **a part** of that assembly line, **a file** that one of the parts loads, or **a knob** on the artist.

2 Modes WHAT YOU HAND IT

The letters only describe what goes **in**. The **VA** on the end means Video + Audio comes out. Every mode makes sound.

MODE	YOU PLUG IN	WHAT H3 DOES WITH IT	REACH FOR IT WHEN
T2VA Text to video	A prompt. Nothing else.	Invents everything: the people, the place, the camera, the sound.	You have no picture to start from. Most freedom, least control over exactly how things look.
I2VA Image to video	A prompt and one picture, as `first_frame`.	Your picture becomes the literal opening frame. The video plays forward from it.	You already have the exact look you want and need it to move and talk.
FL2VA First and last frame	A prompt, a `first_frame` and a `last_frame`.	Opens on picture one, lands on picture two, and invents the trip between them.	The ending has to hit an exact image: a transition, a transformation, a before and after. Describe the trip, not the two pictures.
L2VA Last frame	A prompt and only a `last_frame`.	Invents a beginning and arrives at your picture.	You want a reveal that ends on one specific shot.
Ref2VA Reference to video	A prompt and references: up to 9 pictures, 3 clips, 3 audio clips.	Treats them as examples, not as frames. This is her face. This is his voice. Move the camera like this clip.	You want the same character or voice in a brand new scene.

Two files cover all five

T2VA, I2VA, FL2VA and L2VA are one model file (`fl2va`) and one node, MiniMax H3 Image to Video. Which mode you get depends only on what you plug in: nothing, a first frame, both, or just a last frame. Ref2VA is a different file (`ref2va`) with its own node, MiniMax H3 Reference to Video.

First frame vs reference

A first frame **is** the opening shot, pixel for pixel. A reference is a photo you hand the artist and say: this person, somewhere else. In the prompt you call references by tag, in the order you plugged them in: `<Picture 1>` `<Video 1>` `<Audio 1>`

Faces need to be close

H3 holds a face well in close-ups and loses it in wide shots. If it matters who the person is, frame them close.

3 The parts WHAT EACH FILE YOU LOAD IS FOR

A workflow loads **four model files**. Each one has its own loader box on the canvas.

Diffusion model THE ARTIST

`also called: checkpoint, UNet, DiT | field: unet_name`

WHAT IT DOES

Makes the video and the sound. This **is** the model. Everything else exists to serve it.

WHY YOU CARE

It is the biggest file and it works on every single step, so its size against your 16 GB decides how fast you go. Picking this file is also how you pick the mode: `fl2va` or `ref2va`.

Checkpoint, UNet and DiT all mean this one file. DiT is just the type of network inside. Nothing you do depends on it.

Text encoder THE READER

`also called: CLIP | field: clip_name`

WHAT IT DOES

Reads your prompt, looks at any pictures you connected, and hands the artist a set of numbers that mean the same thing. The artist never sees your words, only those numbers.

WHY YOU CARE

H3's reader is a full language and vision model (Qwen3-VL, 32 billion parameters). That is why long, detailed prompts work, and why it is nearly as big as the artist. It runs once per prompt: change only the seed and ComfyUI reuses its last answer.

Set the loader's type to `minimax`. It is called CLIP because older models used a small reader by that name.

Video VAE THE UNPACKER

`field: vae_name`

WHAT IT DOES

The artist works on a compressed version of the video, 16 times smaller on each side. The VAE turns that back into full frames at the end. It also squeezes your input pictures down at the start.

WHY YOU CARE

Unpacking the whole clip is a burst of memory right at the end. If a run dies after the last step already finished, this is usually where.

Audio VAE THE UNPACKER, FOR SOUND

`field: vae_name`

WHAT IT DOES

The same job for the soundtrack. Turns the artist's compressed sound into 32 kHz stereo audio.

WHY YOU CARE

Barely. It is a small file and never the bottleneck. A workflow just needs both VAEs loaded to give you a video with sound.

Add-ons and words you will keep seeing

LoRA

A small patch file that sits on top of the artist and bends it toward something: a face, a style, a kind of motion, or speed. **Strength** is how hard it bends. 0 is off, 1.0 is what its maker intended, and well past that the picture overcooks. It has to be made for H3. A LoRA for another model does nothing useful here.

RefMod

A community add-on, not part of ComfyUI itself. It is a reference picture or clip that someone already ran through the VAE and saved as a tiny file, so a character loads like a preset and you skip wiring in photos. Nothing was trained. It does the same job as plugging the pictures into Ref2VA. Still experimental.

Checkpoint

Just means a saved model file. Around H3, people use it for the diffusion model.

.safetensors

The container, like .zip or .mp4. It tells you nothing about what is inside. The name is literal: the file can only hold numbers, so it cannot hide a program the way the older .ckpt and .pt files could.

4 File names WHICH ONE TO DOWNLOAD

Read a model's filename left to right and it tells you exactly what you are getting.

`minimax_h3` `_fl2va` `_pruned` `_int8_convrot` `.safetensors`

minimax_h3
The model family. Every file you load has to be made for H3.

fl2va
The job it was trained for. `fl2va` is text, first frame, last frame. `ref2va` is references.

pruned
The slimmed build. If the word is missing, it is the full-size one.

int8_convrot
How its numbers are stored. This is the part that sets the file size.

.safetensors
The container. Ignore it.

The one idea behind all the letters

A model is a giant list of numbers. The letters say **how many bits each number is stored with**. Fewer bits: smaller file, more of it fits on your card, slightly less exact result. Same trade as JPEG quality or MP3 bitrate.

bf16
16 bits per number. The original: full quality, enormous. Not for a 16 GB card.

int8
8 bits per number. Half the size of bf16.

convrot
The trick that makes 8 bits hold up. Before rounding, the numbers get shuffled so a few extreme values cannot wreck the precision of the rest. You set nothing. Read it as "the good int8". It wants an up-to-date ComfyUI whose startup log says `cu130`.

fp8_scaled
The older 8-bit format, same size as int8. Official advice: only use it if `int8_convrot` will not run for you.

w6a8
6-bit weights, 8-bit activations. Weights are the model's stored numbers. Activations are the numbers flowing through it while it works. The smallest official diffusion file. It is the newest format, so update ComfyUI first.

nvfp4
NVIDIA's 4-bit format, the fewest bits of anything here. The official text encoder ships this way and runs on any card. RTX 50 cards like yours can also do the math on it directly.

awq
The method that picked the 4-bit rounding so the numbers that matter most lose the least. It is part of the text encoder's filename. Nothing to set.

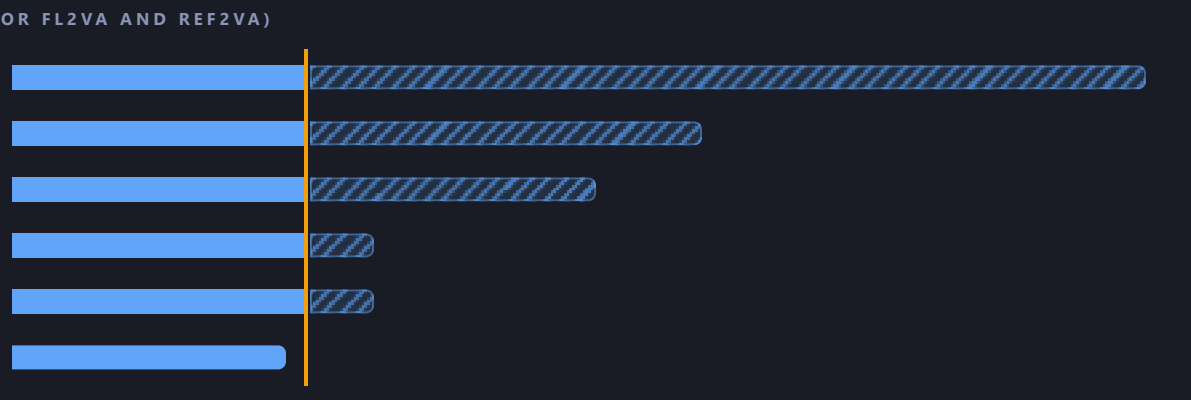
pruned
About 40% smaller. One bulky piece of the model, the part that tracks how far along the cleanup is, was swapped for a compact shared version. ComfyUI's own workflows load this build, and the official advice is to take it whenever both exist.

fp16 / fp32
Plain 16-bit and 32-bit storage. You see them on the VAE files, which are small enough that nobody squeezes them hard.

File size against your 16 GB card

can sit on the card · has to spill over into system RAM · 16 GB: the edge of your card

DIFFUSION MODEL (SAME SIZES FOR FL2VA AND REF2VA)



TEXT ENCODER



Sizes are the official files. In real use less than the full 16 GB is free for the model, because the math itself needs room to work, so treat the line as the best case.

Download this set

PART	FILE	SIZE	GOES IN COMFYUI/MODELS/
Diffusion model text, first and last frame	minimax_h3_fl2va_pruned_int8_convrot.safetensors	19.5 GB	diffusion_models
Diffusion model references, only if you want Ref2VA	minimax_h3_ref2va_pruned_int8_convrot.safetensors	19.5 GB	diffusion_models
Text encoder	qwen3vl_32b_minimax_h3_nvfp4_awq.safetensors	14.6 GB	text_encoders
Video VAE	minimax_h3_video_vae_fp16.safetensors	4.9 GB	vae
Audio VAE	minimax_h3_audio_vae_fp32.safetensors	0.6 GB	vae
Turbo LoRA optional, for fast drafts	minimax_h3_fl2v_turbo_8step_v1.0_comfyui_bf16.safetensors	1.8 GB	loras

This is the exact set ComfyUI's built-in H3 templates ask for. All of it is at huggingface.co/Comfy-Org/MiniMax-H3.

Smaller is not automatically better. A smaller file helps for one reason: less of it spills off the card. Every step down in bits costs a little exactness, so go only as small as you have to. If this set runs out of memory or crawls, the next thing to try is the `pruned_w6a8` diffusion model, then compare it against the default with the same seed.

5 Settings THE KNOBS ON THE ARTIST

Change **one at a time, with the seed fixed**, or you will never know which one did it.

KNOB	WHAT IT ACTUALLY DOES	TURN IT UP AND	START AT
Resolution megapixels, or width x height	How many pixels are in each frame.	Frames get sharper and more detailed. Every step takes longer and needs more memory, climbing at least as fast as the pixel count. Past 1344 x 768 you gain nothing: H3 was trained at about 1 megapixel.	`0.4 MP = 864 x 480` Go up once it runs. 0.98 MP (1344 x 768) is H3's full native size. Skip the 1.0 setting: it lands just over the limit.
Duration seconds, or length in frames	How long the clip is. H3 always makes 24 frames per second, so 5 seconds is 124 frames. The workflow snaps whatever you type to a length the model accepts.	Longer clip, and more work and memory on every single step. H3 was trained on roughly 5 to 15 seconds. Longer than that is untested territory.	`5 seconds` Short clips are the cheapest way to learn what the other knobs do.
FPS frames per second	Not a knob. H3 makes 24 frames for every second of sound, always.	If a save node lets you change it, the same frames just play faster or slower, and the sound drifts out of sync.	`24` Leave it.

Seed	The exact pattern of static the artist starts from. It is a number that picks a pattern, nothing more.	There is no up or down, only different. Same seed with everything else the same gives you the same video again. A new seed is a new take on the same prompt.	`fixed`
			While you test a setting, so the only thing that changed is the thing you changed.
			`random`
			When you just want more takes.

Steps	How many cleanup passes the artist gets.	Cleaner picture, better motion, closer to the prompt. Time rises in a straight line: 20 steps takes twice as long as 10. Most of the gain is in by step 16. Fine detail keeps improving to about 50. Sound finishes later than picture: voices are the weakest part at 8 steps and usable from 12 up.	`20` The official default. Drop to 12 to 16 for quick tests.
Sampler	The method the artist uses on each pass to decide what to clean up next.	Not a more or less knob. A different sampler reaches a slightly different result in the same number of steps.	`res_multistep` The official pick. `euler` is the plain, dependable baseline. Do not tune this early.
Scheduler	The pacing. It decides how much static comes off at each step: big chunks early, when layout and motion get decided, and fine touches late.	Not a more or less knob either. Different pacing, same number of steps.	`simple` For Ref2VA with lots of references, the official notes say `beta` or `normal` often does better.
Shift shift_video, shift_audio	Where the steps get spent. Higher puts more of them on the early stage, where composition and movement get locked in. Lower puts more on late fine detail.	The balance tips toward structure and away from detail. Run a model far from the shift it was built for and the frames can pick up grid patterns.	`12 video, 3 audio` Built in. You only see these if a workflow adds a ModelSamplingMiniMaxH3 node. Leave them.
CFG	In older models: how hard to push toward the prompt, and the thing that makes negative prompts work. It runs the artist twice per step.	Does not apply. H3's official workflows do not have it. They use a Basic Guider: your prompt goes in, with no CFG number and no negative prompt box.	`nothing to set` If a community workflow shows a CFG field, 1.0 means off.
Turbo "Lightning LoRA" in the templates	A LoRA that teaches the artist to finish in 8 steps (or 4) where it normally takes 20.	Runs finish in well under half the time. You pay in softer detail and weaker voices. Set steps to the number in the LoRA's filename, and match it to the model: an `fl2v` turbo for the fl2va file, a `ref2v` turbo for the ref2va file.	`on for drafts, off for keepers` The same seed gives a different video with Turbo on and off, so use it to test prompts, not to preview one exact take.
LoRA strength	How hard a LoRA bends the artist.	Stronger effect, then overcooked: waxy skin, harsh contrast, warped shapes.	`1.0` Then nudge by 0.1 or 0.2.

6 Your hardware WHY IT IS SLOW, AND WHAT THE ERRORS MEAN

H3 is bigger than your graphics card, and that is fine. It runs anyway, just slower. Here is why.

VRAM THE WORKBENCH

The graphics card's own memory. You have 16 GB. Whatever is sitting on it gets worked on at full speed, so this is the number that decides what is comfortable.

RAM THE SHELF

The PC's regular memory. You have 32 GB. Model files wait here when it is not their turn, and anything that does not fit on the workbench lives here.

Offloading THE SHUFFLE

When a model is bigger than the workbench, ComfyUI keeps the overflow on the shelf and swaps pieces on and off the card as each is needed. It is automatic. The video comes out the same. It only costs time.

Your machine, in numbers

The default set is about **40 GB of model**: artist 19.5, reader 14.6, unpackers 5.4. You have 16 GB on the card and 32 GB beside it. It works because the parts **take turns**: the reader goes first and steps aside, the artist does its steps, the unpackers finish. Your tight spot is the 32 GB of RAM as much as the card, so close games and heavy browser tabs before a run. If RAM fills up as well, Windows starts using the disk as memory and everything crawls.

Too slow, or out of memory? Pull these in order

1. **Lower the megapixels.** 0.4, then 0.3. The biggest saving for one click.
2. **Shorten the clip.** Back to 5 seconds.
3. **Fewer steps.** 16, or Turbo at 8. This saves time, not memory.
4. **Smaller files.** The `pruned_w6a8` diffusion model, and `minimax_h3_video_vae_int8_convrot` (2.6 GB) in place of the fp16 video VAE.
5. **Free the machine.** Close everything else that uses RAM or the graphics card.
6. **Sage Attention, last.** An optional speed-up you install yourself. The official docs say it can roughly double speed. If clip endings start morphing or on-screen text turns to garbage, turn it back off.

What an error is telling you

`Red boxes when a workflow opens`
Missing custom nodes, or the workflow needs a newer ComfyUI. Update first, then let ComfyUI Manager install what is missing.

`"Value not in list" on a loader`
That file is not in the right models folder, or it has a different name than the workflow expects. Pick yours from the dropdown.

`"CUDA out of memory" / "Allocation on device"`
The card ran out of VRAM. Levers 1, 2 and 4.

`Dies after the last step finished`
Usually the video VAE running out while unpacking. Lower megapixels or a shorter clip.

`The whole PC crawls or freezes`
RAM is full and Windows is using the disk. Close other programs, or move to smaller files.

The other words in this corner

CUDA / PyTorch THE PLUMBING

PyTorch is the math engine ComfyUI is built on. CUDA is NVIDIA's layer that lets PyTorch use the graphics card. Both come bundled with ComfyUI and you never touch them. You only meet them in messages: "CUDA out of memory" means the card is full, and `cu130` in the startup log means PyTorch built for CUDA 13.0, which is what the int8_convrot files want.

Custom node AN ADD-ON

Every box on the canvas is a node. Custom nodes are boxes written by other people, installed through ComfyUI Manager. **H3 itself needs none**: its nodes are built into ComfyUI. You only need a pack when a workflow you downloaded uses one. Each pack is code running on your PC, so stick to well-known ones.

rgthree A POPULAR PACK

A set of convenience nodes that shows up in a lot of shared workflows. The two you will meet most: a LoRA loader that stacks several LoRAs in one box, and switches that turn whole groups of a workflow on and off.

7 First run A KNOWN-GOOD STARTING POINT

Get one boring clip working with the official setup before you touch anything clever.

The recipe

1. **Update ComfyUI.** H3 support is recent and still moving.
2. **Open the built-in template.** Workflow menu, Browse Templates, search for MiniMax H3, pick text to video.
3. **Get the files.** ComfyUI lists what is missing and where it goes. It is the set from section 4.
4. **Set it small.** `0.4` megapixels, `5` seconds, Lightning LoRA off, which gives 20 steps. Leave sampler and scheduler alone.
5. **Run it.** The first run is the slowest, because every file has to come off the disk. Expect minutes, not seconds.
6. **Then change one thing.** Fix the seed, change a single knob, compare.

The prompt

H3 makes picture and sound together, so the prompt has to cover both. In one block, say **what is seen**, **what is said** and **what is heard**: the look, the shots and camera moves, who does what, the exact words spoken, the background sound, the music.

Long and specific beats short and vague. The reader is a full language model and it uses the detail.

With Ref2VA, say which reference does which job: `<Picture 1>` is the woman, `<Audio 1>` is her voice. The official templates warn that Ref2VA is very sensitive to that wording.

The full official prompt-writing guides are in the docs folder at huggingface.co/MiniMaxAI/MiniMax-H3.

Checked against: ComfyUI H3 docs, Comfy-Org/MiniMax-H3 file list, MiniMax H3 model card, ComfyUI's built-in H3 nodes and templates. October 2026.